



Figure 7: Setting up the IoT sensor

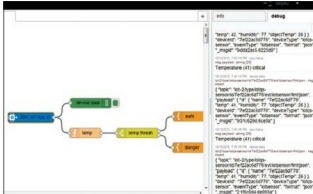


Figure 8: Analysing and logging the status of the sensor on changing the temperature



Figure 9: OpenShift login panel

You can get a view of the temperature sensor simulator at <https://quickstart.internetofthings.ibmcloud.com/iotsensor/>.

In Figure 7, the device ID is mapped with the actual device. Let's assume there is a particular machine and we want to sense its temperature using IoT. In this case, the unique machine ID is mentioned in Node RED so that the application can sense the live temperature or any other required parameter.

In Figure 8, the live logging and analysis of the temperature can be viewed. Suppose the machine under analysis is moved beyond the threshold temperature, the IoT application will sense and display the appropriate message mentioned in the code.

Creating PHP applications in IBM Bluemix

IBM Bluemix provides a PHP starter application template to

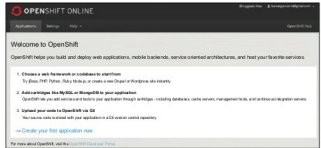


Figure 10: Creating the first application panel in the OpenShift platform



Figure 11: Selecting the PHP platform in OpenShift temperature

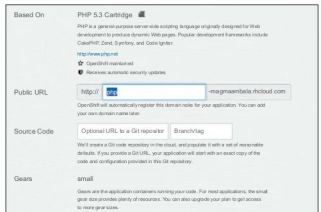


Figure 12: Setting up the URL and related parameters of the app in OpenShift

cloud users so that their own code can be added and pushed back to the cloud server. This starter application template is the boilerplate for PHP Web application development in Bluemix.

Initially, a new PHP starter application is activated and starter code is downloaded. After that, the App name and host are provided by the user. For further implementation, the `cf` (Cloud Foundry) command line tool is used. `cf` is the CLI (command line interface) that can be used to deploy and control the applications on Bluemix.

To log in to the Bluemix environment, type:

```
$ cf login -a https://api.ng.bluemix.net -o <myor name> -s <myspacename>
```

After making changes to the code, deploy the application again. For deploying the modified application to Bluemix, the `cf` command is used: